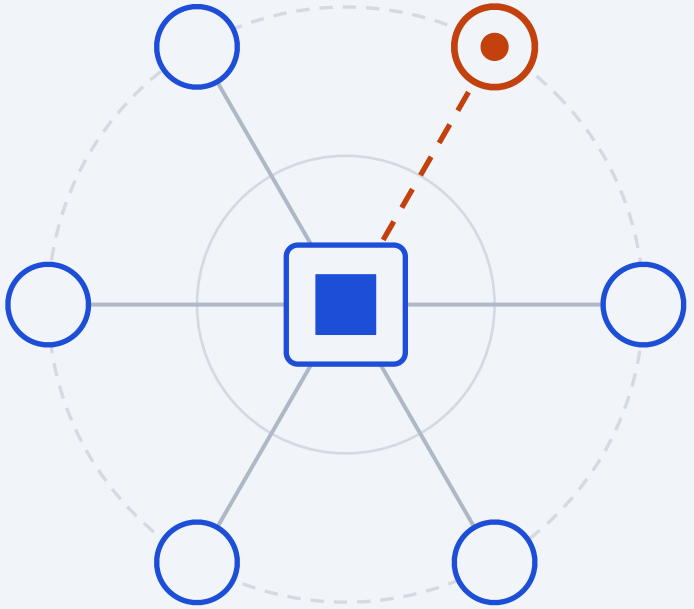
PROJECT PROPOSAL · CSCI 6330

Federated Intrusion Detection Under Label-Flipping Poisoning



Five honest offices, one poisoned (orange)

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01 — THE PROBLEM

Share models, not data

1 TRAIN

Train locally

Each office keeps its raw traffic and trains a small detector.

2 SHARE

Share updates

Only model updates go to the server, never the data.

3 COMBINE

Combine

The server averages them and sends the better model back.

! RISK

One compromised office can corrupt the shared model.

02 — OUR QUESTION

How much detection performance is lost when one office sends flipped labels, and does a robust average recover it?

THE ATTACK

1 of 6 offices

Relabels 50% of its attack samples as benign every round. Everyone else is honest.

WHAT WE EXPECT

F1 drops by over 3 points

Trimmed-mean averaging recovers at least half. Smaller effects would prove us wrong.

03 — THE DATA

Our data: CICIoT2023

DATASET	**CICIoT2023, network flows from IoT devices**
FLOWS	**~300,000 in our subset**
CLASSES	**7 attack families + benign traffic**
SPLIT	**20% test · 10% validation · 70% offices**
OFFICES	**6 simulated, built with k-means**

THE SEVEN ATTACK FAMILIES

DDoS · DoS · Recon · Web-based · Brute force · Spoofing · Mirai

Note: the offices are simulated from one lab testbed. No real-world claims.

HOW OFFICES ARE BUILT

1. **k-means** groups similar traffic. No labels, IPs or timestamps.
2. The **test set** is held out first and never used for tuning.
3. An **even split** of the same data is the control.

04 — THE PLAN

How we will do the project

MILESTONE 1

Oct 25

Prepare and baseline

Build the subset and six offices. Train the centralized model.

MILESTONE 2

Nov 15

Federate and attack

Train clean federated models. Then one office flips its labels.

FINAL TALK

Nov 30

Defend

Test trimmed-mean averaging, with and without the attacker.

FINAL REPORT

Dec 7

Write up

Report, limitations, and code with a README.

HOW WE MEASURE

Macro F1, missed attacks per class, results by round and office, and three runs.

STAYING SAFE

Simulated only, on static public data, with a clean baseline for every attack.

05 — Q AND A

Questions?

